



Quantum-based Tensor Contraction Layers

A Hybrid Classical–Quantum Prediction Head

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Motivation: The Deployment Challenge

RESOURCE-CONSTRAINED ML

- ▲ Edge devices, federated nodes, all impose **tight memory & compute budgets**.
- ▲ Key considerations: memory, communication, and latency costs.

TWO COMPRESSION AXES

- ▲ **Backbone compression** (pruning, quantization, low-rank conv) — reduces feature-extractor cost.
- ▲ **Head compression** (TCL [1]) — prevents the prediction head from re-introducing a parameter bottleneck.

VGG-19 FC head (ImageNet)

Feature map: $7 \times 7 \times 512$

FC1 ($\rightarrow 4096$): 102.8 M

FC2 ($\rightarrow 4096$): 16.8 M

FC3 ($\rightarrow 1000$): 4.1 M

Total head: 123.6 M

FC1 alone = $7^2 \times 512 \times 4096$.

Motivation: From FC to TCL to QTCL

Head	Formula	Example	Reduction
FC	$d^2 D^2 OC$	102.8 M	—
TCL [1]	$2dD + OC$	2.10 M	$\sim 49\times$
QTCL (ours)	$2dD + n_q(C+O) + P_\theta$	36.9 K	$\sim 57\times$ (TCL)

VGG-19 FC1: $D=7$, $d=1$, $C=512$, $O=4096$, $n_q=8$.

- ▲ TCL compresses the spatial modes well, but the **channel term** OC still dominates.
- ▲ QTCL replaces OC with $n_q(C+O)$: a **product becomes a sum**.
- ▲ The PQC overhead $P_\theta = \mathcal{O}(n_q L)$ is **input-size-independent**.

Channel cost

$$OC = 2,097,152$$

↓

$$n_q(C+O) = 36,864$$

$\approx 57\times$ further reduction

Background: Fully Connected Head

STANDARD APPROACH

- ▲ Final conv stage returns $\mathcal{X} \in \mathbb{R}^{D \times D \times C}$.
- ▲ Flatten to $\mathbf{x} \in \mathbb{R}^{D^2 C}$, apply dense matrix:

$$\mathbf{y} = \mathbf{V}\mathbf{x}, \quad \mathbf{V} \in \mathbb{R}^{H \times D^2 C}.$$

- ▲ Cost: $H \cdot D^2 C$ parameters.

PROBLEMS AT SCALE

- ▲ Flattening **destroys the multiway structure** of the activation tensor.
- ▲ Parameter count grows as a **product of all dimensions** (H, D^2, C).

Key question

Can we act *directly* on the activation tensor $\mathcal{X} \in \mathbb{R}^{D \times D \times C}$ with **mode-wise projections**, preserving structure and slashing the parameter count?

Layer	Params
FC	$d^2 D^2 OC$
TCL [1]	$2dD + OC$

Background: Tensor Contraction Layer (TCL)

ACTING DIRECTLY ON THE TENSOR [1]

- ▲ Avoids flattening — contracts each mode separately:

$$\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3$$

with $\mathbf{U}_1, \mathbf{U}_2 \in \mathbb{R}^{d \times D}$ (spatial), $\mathbf{U}_3 \in \mathbb{R}^{O \times C}$.

- ▲ Total cost: $2dD + OC$ — a massive reduction.
- ▲ Preserves multiway structure throughout.

RESIDUAL BOTTLENECK

- ▲ Channel mode params dominates the spatial mode when $C, O \gg d, D$ (e.g., $OC = 2,097,152, C = 512, O = 4096$).

Mode- n product

$\mathcal{Y} = \mathcal{X} \times_n \mathbf{U}$ contracts the n -th mode of $\mathcal{X}$ with matrix $\mathbf{U}$, reducing the corresponding dimension from I_n to the row count of $\mathbf{U}$ — while keeping all other modes intact.

Remaining gap

TCL provides a $\sim 49\times$ reduction over FC, yet the channel factor $\mathbf{U}_3$ alone still holds **2.1M params**.

QTCL targets precisely this term.

Method: Quantum-based Tensor Contraction Layers

REPLACE THE CHANNEL FACTOR

Swap the dense $\mathbf{U}_3 \in \mathbb{R}^{O \times C}$ with a **blended classical-quantum map**:

$$\mathbf{W} = \mathbf{A}(\mathbf{I} + \Phi_\theta)\mathbf{B}$$

- ▲ $\mathbf{B} \in \mathbb{R}^{n_q \times C}$ — project C channels **down** to n_q .
- ▲ $\Phi_\theta \in \mathbb{R}^{n_q \times n_q}$ — **PQC unitary mixing** in the bottleneck.
- ▲ $\mathbf{A} \in \mathbb{R}^{O \times n_q}$ — project **up** to O classes.
- ▲ $\mathbf{I}$ residual $\Rightarrow$ stable initialization (scaled low-rank start).

Channel cost

$$OC \longrightarrow n_q(C+O) + P_\theta$$

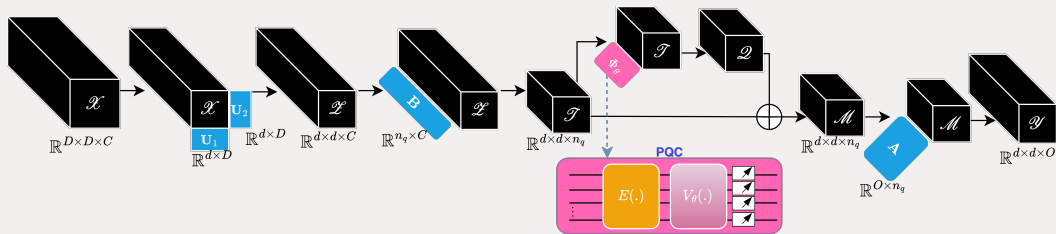
$a \times$ becomes $a +$.

Parameter count

$$\underbrace{2dD}_{\text{spatial}} + \underbrace{On_q + n_q C}_{\text{classical channel}} + \underbrace{P_\theta}_{\text{PQC}}$$

QTCL and TCL share the *same* spatial budget ($2dD$), so all accuracy differences are attributable entirely to PQC mixing.

Method: QTCL Architecture



Quantum-based Tensor Contraction Layer

Spatial (TCL-inherited)	Channel bottleneck	Quantum mixing
$\mathbf{U}_1, \mathbf{U}_2$ unchanged from TCL	n_q -dimensional latent space	PQC Φ_θ on n_q qubits

Method: End-to-End Gradients

INTERMEDIATES

$$\mathbf{Z} = \mathbf{X} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2, \quad \mathbf{S} = \mathbf{I} + \Phi_\theta, \quad \mathbf{W} = \mathbf{A}\mathbf{S}\mathbf{B}.$$

CLASSICAL GRADIENTS (CHAIN RULE)

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{G}_{(3)} \mathbf{Z}_{(3)}^\top$$
$$\frac{\partial \mathcal{L}}{\partial \mathbf{A}} = \frac{\partial \mathcal{L}}{\partial \mathbf{W}} (\mathbf{S}\mathbf{B})^\top, \quad \frac{\partial \mathcal{L}}{\partial \mathbf{B}} = \mathbf{S}^\top \mathbf{A}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{W}}.$$

PQC ANGLES — PARAMETER-SHIFT RULE [3]

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{1}{2} [\mathcal{L}(\theta_k + \frac{\pi}{2}) - \mathcal{L}(\theta_k - \frac{\pi}{2})]$$

propagated via $\frac{\partial \mathcal{L}}{\partial \mathbf{S}} = \mathbf{A}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{W}} \mathbf{B}^\top.$

Fully differentiable

All QTCL parameters — classical weights **and** PQC angles — are updated end-to-end via standard backprop + parameter-shift rule.

No custom gradient approximations needed.

Isolated PQC contribution

QTCL and TCL share the exact same spatial budget ($2dD$).

Any accuracy difference is **entirely attributable to PQC mixing** in the channel bottleneck.

Method: Complexity & Parameter Count

Parameter count

$$\# \text{param}_{\text{QTCL}} = \underbrace{2dD}_{\text{spatial (same as TCL)}} + \underbrace{On_q + n_q C}_{\text{channel (classical)}} + \underbrace{P_\theta}_{\text{PQC angles}}$$

Method	Forward	Backward	#Params
FC	$\mathcal{O}(d^2 D^2 OC)$	$\mathcal{O}(d^2 D^2 OC)$	$d^2 D^2 OC$
TCL	$\mathcal{O}(dD^2 C + d^2 DC + d^2 OC)$	$\mathcal{O}(dD^2 C + d^2 DC + d^2 OC + d^2 OD + DdOC)$	$2dD + OC$
QTCL	$\mathcal{O}(\dots + n_q^2 + n_q^2 C + On_q C + SL \text{ poly}(n_q))$	$\mathcal{O}(\dots + P_\theta SL \text{ poly}(n_q))$	$2dD + n_q(C + O) + P_\theta$

S = parameter-shift shots; L = circuit depth. QTCL adds only an **input-size-independent** quantum overhead over TCL.

Experimental Settings

DATASET

- ▲ **CIFAR-100**: 60k 32×32 images, 100 classes (50k train / 10k test)
- ▲ Standard augmentation: random crop + horizontal flip

BACKBONES & VARIANTS

- ▲ **VGG-19** and **AlexNet**
- ▲ Suffix **1**: replace FC1 only moderate
- ▲ Suffix **12**: replace FC1+FC2 aggressive
- ▲ Baselines: **TCL1/TCL12** (classical channel)
- ▲ Proposed: **QTCL1/QTCL12** (quantum-mixed channel)

TRAINING PROTOCOL

- ▲ SGD: $\eta=0.01$, momentum 0.9, weight decay 10^{-4}
- ▲ 160 epochs, batch 128, StepLR ($\gamma=0.1$)

METRICS

- ▲ Top-1 validation accuracy
- ▲ Head space saving: $1 - n_{\text{mod}}/n_{\text{orig}}$
- ▲ ΔAcc vs. uncompressed backbone

PQC configuration

Hardware-efficient ansatz (linear entanglement)
 $n_q = 8$ qubits, $L = 2$ layers
 $P_\theta = 16$ trainable angles
Simulator: PennyLane (analytic statevector)

Results: AlexNet / CIFAR-100

MODERATE COMPRESSION

Model	Acc (%)	Δ Acc	Saving
AlexNet	65.70	—	—
TCL1	66.56	+0.86	62.77%
QTCL1	63.53	-2.17	63.00%

AGGRESSIVE COMPRESSION

Model	Acc (%)	Δ Acc	Saving
AlexNet	65.70	—	—
TCL12	65.69	-0.01	98.64%
QTCL12	57.92	-7.78	99.10%

Moderate regime

TCL1 slightly **surpasses** the baseline (+0.86 pp) at 62.77% savings — consistent with AlexNet being overparameterized for CIFAR-100.

Aggressive regime

TCL12 holds near-baseline at 98.64% savings — a strong classical result.

QTCL12 trades accuracy for **ultra-extreme compression** (99.10%), attractive in the most memory-constrained regimes.

Takeaway

AlexNet's **shallow, sparse features** limit quantum mixing benefit. Wider/deeper PQCs may help for shallower backbones.

Results: VGG-19 / CIFAR-100

MODERATE COMPRESSION

Model	Acc (%)	Δ Acc	Saving
VGG-19	72.21	—	—
TCL1	71.83	-0.38	45.80%
QTCL1	71.47	-0.74	46.50%

AGGRESSIVE COMPRESSION

Model	Acc (%)	Δ Acc	Saving
VGG-19	72.21	—	—
TCL12	67.48	-4.73	97.27%
QTCL12	71.67	-0.54	98.68%

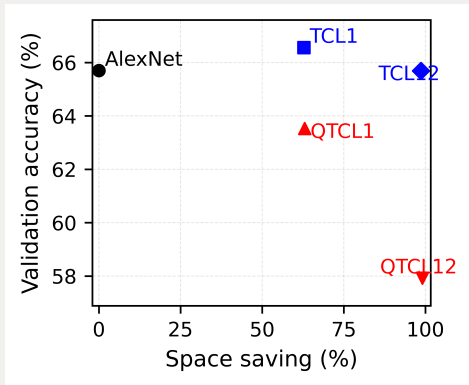
Moderate regime

QTCL1 and TCL1 are **broadly competitive** at similar savings ($\approx 46\%$), both within 0.74 pp of the full baseline.

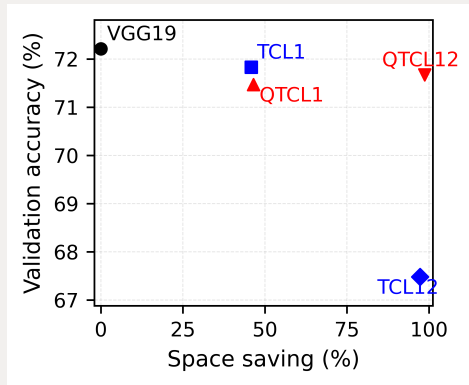
Headline: QTCL12

71.67% Top-1 accuracy
at **98.68%** head-parameter savings
+4.19 pp over TCL12 (67.48%)
Closes nearly all gaps to baseline
(72.21%)

Results: Accuracy–Compression Pareto



AlexNet



VGG-19

Key observation

Ablation: Quantum Width n_q and Depth L

Top-1 accuracy (%), QTCL12 on VGG19

L	$n_q = 4$	$n_q = 8$
1	66.19	71.58
2	65.60	71.67
3	66.14	71.51

Trend 1 — Width dominates depth

$n_q = 8$ beats $n_q = 4$ by **5–6 pp** across all depths. The bottleneck dimension n_q is the primary driver of performance.

Trend 2 — Depth saturates quickly

Within $n_q = 8$, accuracy is stable: **71.51–71.67%** for $L \in \{1, 2, 3\}$.

Design implication

Prioritize **wider** qubit registers over deeper circuits when scaling QTCL to harder tasks.

Chosen default

$$n_q = 8, \quad L = 2$$

16 trainable PQC angles

Since QTCL and TCL share the same spatial budget ($2dD$), these differences isolate the pure effect of PQC width and depth.

Conclusion

WHAT WE PROPOSED

- ▲ **QTCL**: replaces TCL's channel factor with $\mathbf{W} = \mathbf{A}(\mathbf{I} + \Phi_\theta)\mathbf{B}$.
- ▲ Channel cost: **product** $OC \rightarrow \text{sum } n_q(C+O)$
- ▲ PQC unitary mixing: **implicit regularization** at high compression.
- ▲ Fully end-to-end differentiable via the parameter-shift rule

KEY RESULT — VGG-19 / CIFAR-100

- ▲ **71.67%** Top-1 at **98.68%** head-parameter savings.
- ▲ **+4.19 pp** over TCL12 at comparable compression.
- ▲ Closes nearly all gaps to FC baseline, 72.21%.

Broader applicability

Drop-in for any FC/TCL head · no backbone modifications ·
 $n_q=8$, $L=2$ compatible with near-term quantum hardware.

Limitations

AlexNet benefit limited at $n_q=8$.
CIFAR-100 only; distribution-shift robustness open.
Simulation cost scales with n_q (statevector).

Future work

Test/Analyze/improve training stability and barren plateau phenomenon.
Robustness on **CIFAR-100-C**.
Real hardware + noise-aware training.

Code: github.com/vantnprof/QTCL

References

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Questions?

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